

# James Philip Pulver

## Senior Data Architect & Engineering Leader

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### Profile

Senior **Data Architect & Engineering Leader** with 15+ years of experience designing, building, and optimizing large-scale, cloud-native data platforms. Expert in **AWS, Azure, SQL Server, Databricks, and Snowflake**, with deep skills in modern **lakehouse architectures, ETL/ELT pipelines, and real-time data streaming**. Proven record of **leading cross-functional teams**, implementing **data governance frameworks**, and enabling **BI & analytics adoption** across healthcare, finance, and enterprise technology. Recognized as a **hands-on leader** who drives innovation, mentors engineering talent, and delivers high-impact solutions that improve compliance, performance, and business outcomes.

### Professional Experience

**Amazon Web Services, Staff Data Engineer / Data Architect**

12/2021 — Present | Seattle, WA

- **Architected and deployed large-scale data pipelines** leveraging **PySpark, Databricks, AWS S3, and AWS Glue**, enabling both batch and near real-time ingestion of **5+ TB of data per day** across global enterprise systems.
- **Designed and implemented a modern lakehouse architecture** using **Delta Lake, Amazon Redshift, and Snowflake**, delivering scalable, secure, and cost-efficient analytics capabilities for internal and external stakeholders.
- **Built modular, reusable ETL frameworks** with **Apache Airflow, AWS Lambda, and Azure Data Factory (ADF)**, streamlining ingestion of APIs, relational databases, and streaming data sources, while reducing development cycle time by **40%**.
- **Developed and optimized streaming data solutions** integrating **Apache Kafka and Amazon Kinesis**, supporting time-sensitive analytics for fraud detection, operational monitoring, and customer experience insights.
- **Tuned Spark-based ETL workloads** by implementing **efficient partitioning strategies, job parallelism, and cluster auto-scaling**, resulting in faster execution and reduced processing costs.
- **Established robust data quality and governance frameworks** with **Great Expectations, Glue Data Catalog, and metadata-driven validation**, improving trust in analytics and ensuring compliance with enterprise standards.
- **Implemented CI/CD automation pipelines** using **Terraform, GitHub Actions, and AWS CodePipeline**, enabling continuous integration, automated testing, and multi-environment deployment of data pipelines and infrastructure-as-code.
- **Collaborated with product, operations, and finance teams** to design **dashboards and KPI-driven analytics**, delivering insights that improved cloud revenue optimization and cost management.
- **Mentored and coached global data engineering teams**, leading **architecture design reviews, code quality standards, and cloud-native engineering best practices** across distributed teams.
- **Partnered with security and compliance teams** to enforce **IAM roles, KMS encryption, and data masking policies**, ensuring adherence to **HIPAA, GDPR, and SOC 2** requirements across enterprise data platforms.
- **Led cloud cost-optimization initiatives** by implementing **storage lifecycle policies, intelligent tiering, and query performance tuning**, reducing AWS data infrastructure spend by **25%**.

- **Directed and managed a cross-functional BI and data engineering team**, overseeing the design, development, and delivery of enterprise reporting and analytics solutions across clinical, operational, and financial domains.
- **Modernized legacy reporting systems** into a centralized, cloud-ready **SQL Server Data Warehouse**, integrating with **Azure SQL Database** and improving query performance by **45%**, reducing infrastructure costs by **20%**, and enabling analytics scalability.
- **Architected scalable ETL pipelines** using **SSIS, Python, T-SQL, and Azure Data Factory** to integrate **EHR/EMR (Epic, Cerner), claims, HR, and financial systems**, ensuring timely, accurate, and trusted reporting for executives and clinicians.
- **Developed healthcare-specific data marts and OLAP models** leveraging **SQL Server Analysis Services (SSAS) and Power BI**, enabling self-service analytics that cut reporting turnaround from days to hours for leadership and department heads.
- **Implemented robust data governance frameworks** including **metadata management, HIPAA-compliant role-based security, and data cataloging tools (Collibra, Azure Purview)**, improving audit readiness and reducing regulatory risks.
- **Automated financial, clinical, and operational dashboards** in **Power BI, SSRS, and Tableau**, reducing manual reporting by **60%** and saving ~400 staff hours annually across multiple departments.
- **Partnered with clinicians and finance executives** to deliver KPI-driven analytics for readmission reduction, patient flow optimization, and cost control, contributing to a **12% drop in patient readmissions** and measurable improvements in hospital efficiency.
- **Optimized SQL queries, ETL jobs, and nightly refresh cycles**, reducing processing time from **8+ hours to under 3 hours**, ensuring near real-time data availability for mission-critical decision-making.
- **Delivered clinical and financial analytics** that reduced patient readmissions by 12% and improved operational efficiency.
- **Championed BI adoption** by training **100+ users**, building reusable **Power BI and SSRS dashboards**, and enabling a data-driven culture across 15+ departments, increasing BI adoption by **70%**.
- **Mentored and coached junior engineers and analysts**, leading code reviews, establishing engineering best practices, and supporting career development, which helped maintain a **95% retention rate** within the BI & analytics team.
- **Served as a strategic advisor on enterprise data initiatives**, aligning hospital-wide **data strategy with cloud adoption (Azure), advanced analytics, and compliance priorities** to support patient care, financial sustainability, and operational excellence.

- **Designed and implemented ETL pipelines** using **SQL Server Integration Services (SSIS) and SQL Server**, consolidating pharmacy claims, patient, and provider data into a unified reporting environment.
- **Developed and optimized data warehouse models** to support **clinical and operational analytics**, enabling faster decision-making across pharmacy operations and patient care management.
- **Built interactive dashboards and reports** that provided **leadership** with **real-time visibility** into key trends, program utilization, and resource allocation, enabling faster and more **informed decision-making**.
- **Automated recurring financial, compliance, and operations reports**, saving **~25 staff hours per week** and ensuring consistent, accurate delivery of critical reports.
- **Improved SQL performance** with **index tuning, partitioning, and stored procedure optimization**, reducing query execution times by **up to 60%** on high-volume claims data.
- **Collaborated with business stakeholders** to translate pharmacy and clinical requirements into scalable **BI solutions**, ensuring alignment with compliance and regulatory standards.
- **Contributed to data governance efforts** by standardizing data definitions and implementing validation checks, which increased **trust and adoption of BI reporting** across teams.

**PCV Murcor**, *Senior Data Engineer / SQL Developer*

09/2012 – 03/2013 | Pomona, CA

- **Developed and maintained SQL Server databases and ETL processes** to support appraisal and real estate data operations, ensuring timely delivery of accurate and consistent reporting to business stakeholders.
- **Created automated stored procedures and SQL scripts** to streamline **data extraction, transformation, and loading (ETL)**, reducing manual intervention by **30%** and improving reliability of recurring data processes.
- **Optimized SQL queries, indexing strategies, and schema design**, which cut report runtimes by up to **50%** and enhanced overall system performance for high-volume appraisal datasets.
- **Designed and deployed reporting solutions** using **SSRS**, providing finance, compliance, and client service teams with real-time access to operational metrics and loan pipeline visibility.
- **Partnered with business analysts and end users** to translate functional requirements into actionable **BI reporting solutions**, increasing transparency into loan and property valuation pipelines.
- **Implemented data validation checks and auditing processes**, improving quality control and ensuring compliance with **client-specific SLAs and regulatory standards**.

**County of Orange – Social Services Agency**,

06/2010 – 08/2012 | Santa Ana, CA

*Senior Data Engineer / SQL Developer*

- **Designed and developed SQL Server databases, ETL processes, and reporting solutions** to support county-wide **social service programs**, including welfare, child support, and community health initiatives.
- **Built automated data pipelines** for integrating **case management, eligibility, and benefits data** across multiple systems, improving accuracy and timeliness of reporting.
- **Created dashboards and analytical reports in SSRS** that provided insights into **caseload trends, program utilization, and resource allocation**, driving more data-informed policy decisions.
- **Optimized SQL queries, indexing strategies, and stored procedures**, reducing **report runtimes by 40%** and enhancing performance under heavy workloads.
- **Partnered with program managers and compliance officers** to define requirements, ensuring reports met **state and federal regulations** for accuracy, timeliness, and transparency.
- **Implemented data validation and auditing processes** to improve **data quality**, reducing reporting errors and increasing trust in agency analytics.
- **Supported ad-hoc reporting requests and emergency analyses** for **executive leadership**, delivering critical insights during funding reviews and operational audits.
- **Collaborated cross-functionally with IT, analysts, and program staff**, translating **business needs into technical solutions** and ensuring adoption of BI tools and processes.

**Precision Discovery**, *Data Engineer / Analytics Developer*

07/2007 – 05/2010 | Irvine, CA

- **Developed and maintained SQL Server databases** supporting eDiscovery and litigation data workflows, ensuring high performance, scalability, and data accuracy under strict compliance requirements.
- **Built and optimized ETL processes** using **SSIS and custom SQL procedures**, automating ingestion of large document collections, case metadata, and structured/unstructured data.
- **Designed interactive reporting dashboards in SSRS and Excel/VBA**, delivering real-time visibility into evidence tracking, billing metrics, and case progress for legal teams.
- **Automated repetitive data preparation tasks** with **VBA macros and scripts**, reducing manual effort for analysts by **30%** and increasing processing efficiency.
- **Implemented indexing, partitioning, and performance tuning strategies** to optimize query execution and reduce ETL runtimes by **40%**, ensuring data was available on tight litigation deadlines.
- **Collaborated with attorneys and compliance staff** to translate business requirements into data solutions, aligning technical designs with case strategy and regulatory obligations.
- **Developed custom validation and audit routines**, improving data integrity and reducing downstream reporting errors by **25%**.
- **Contributed to tool evaluations and pilots** for advanced eDiscovery and analytics software, helping the team adopt more efficient platforms for document review and analysis.
- **Provided technical guidance to junior developers**, sharing SQL best practices and process automation techniques, strengthening overall team capability.

## **Murphy and Harrison,**

08/2006 – 10/2007 | Claremont, CA

*Database Developer / Data Analyst (VBA & SQL)*

- **Developed and maintained SQL Server databases** supporting internal business applications, ensuring accurate and consistent data for finance and operations teams.
- **Wrote complex SQL queries, stored procedures, and functions** to generate reports and automate recurring data processes, improving reporting efficiency by 30%.
- **Built Excel-based reporting solutions with VBA macros**, automating manual workflows and reducing turnaround time for financial and operational reports.
- **Created ETL scripts** to cleanse and transform raw data, enabling standardized reporting across multiple departments.
- **Optimized SQL query performance** by implementing indexing strategies and query tuning, reducing execution times by up to 20–25%.
- **Collaborated with cross-functional teams** to gather reporting requirements and translate business needs into technical database solutions.
- **Delivered ad-hoc reports and dashboards** that provided leadership with visibility into key KPIs and supported faster decision-making.
- **Documented database schemas, ETL processes, and reporting workflows**, improving maintainability and enabling smoother handoffs to future developers.

## **Skills**

### **Data Engineering & Architecture**

- Data modeling, schema design, data warehouse & lakehouse architectures (SQL Server, Redshift, Snowflake, Delta Lake)
- Batch and real-time ETL pipelines with SSIS, Azure Data Factory, AWS Glue, Databricks (PySpark), Apache Airflow

### **Programming & Automation**

- Python (pandas, PySpark, automation scripts)
- Advanced SQL / T-SQL (stored procedures, triggers, query optimization, indexing, partitioning)
- VBA macros for Excel-based automation & reporting

### **Data Governance & Compliance**

- Metadata management, data cataloging (Collibra, Azure Purview, Glue Data Catalog)
- Data quality frameworks (Great Expectations, custom validations, audit checks)
- Regulatory compliance: HIPAA, GDPR, SOC 2, role-based security, encryption & masking

### **Data Analysis & Business Intelligence**

- Dashboard design and self-service analytics using Power BI, Tableau, SSRS
- KPI monitoring, trend analysis, operational and financial reporting
- Ad-hoc and executive reporting, providing actionable insights for decision-making

### **Cloud Platforms & Infrastructure**

- AWS: S3, Lambda, Kinesis, Glue, Redshift, IAM, CodePipeline
- Azure: SQL Database, Data Factory, Purview, Synapse
- Infrastructure as Code: Terraform, GitHub Actions, CI/CD automation

### **Leadership & Strategy**

- Leading cross-functional teams, code reviews, engineering best practices
- Mentorship & career development for analysts/engineers

## **Education**

**Master of Science (MS), Information Systems,**  
*California State University, Fullerton*

08/2010 – 05/2012 | Fullerton, CA

**Bachelor of Arts (BA), Sociology,**  
*University of California, Los Angeles (UCLA)*

09/2002 – 06/2006 | Los Angeles, CA